A Pre-Deployment & Post Diagnostic Framework for Generalisation Failure in Clinical Prediction Models

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Abstract—**Recent studies highlight a common problem in clinical machine learning: models that perform well during development often do not generalize to new patient groups (1; 2). This problem is seen across areas such as psychiatry, genomics, and biomarker research. For example, Chekroud and colleagues found that the accuracy of schizophrenia treatment models dropped from 0.72 during internal testing to 0.54 on separate datasets (3). This drop in performance is usually linked to issues such as changes in data, misleading connections, hidden factors, and evaluation errors (2; 4).**

Although new methods like invariant risk minimization and causal modeling offer theoretical fixes, they are rarely used in practice(5). These methods usually require strong assumptions, such as data from multiple training settings or detailed cause-and-effect models, which are rarely available in real clinical settings. Because of this, experts are good at explaining why models fail after the fact but still lack useful tools to predict these problems before using the models (1).

To fill this gap, a practical three-step process for checking risks before deployment is suggested. First, differences between the training data and the target patient groups are measured using tools such as the Maximum Mean Discrepancy, Kullback–Leibler divergence, and Wasserstein distance (6; 7; 8). Second, the stability of individual features is tested across different settings using cross-context permutation importance to find unreliable predictors. Third, these results guide clear decisions about deployment, such as whether to deploy, adjust, limit, or hold back the model, supported by data-based limits and ongoing monitoring after deployment.

The objective is to deliver a practical, accessible pipeline that clinical machine learning teams can implement using standard tools, without specialized expertise in causal inference or complex modeling assumptions.

Index Terms—**dataset shift, generalisation, clinical prediction models, distributional divergence, feature stability, deployment diagnostics, machine learning, monitoring**

I Introduction

Each year, thousands of machine learning models are developed to predict clinical outcomes, such as cardiovascular risk and treatment response, using electronic health records, genomic data, and imaging(3). The premise is that patterns in patient data will improve decision-making for future cases. This promise has attracted substantial funding, regulatory scrutiny, and clinical interest.

However, most models are evaluated with methods that exaggerate their real-world effectiveness by testing only on data similar to what the model has seen. These approaches do not address how models fare across hospitals, regions, or time periods. Evidence shows substantial performance drops in such settings: for instance, Chekroud et al. (3) reported a decline from 0.72 within trial to 0.54 externally in schizophrenia models, and comparable failures occur across fields from genomics to molecular property prediction(2; 5; 9; 10)

These failures have tangible consequences. A model that performs well during development but fails at deployment may lead to patient misclassification, denial of beneficial treatments, or exposure to inappropriate interventions. Unlike pharmaceuticals, prediction models can be adopted based only on positive cross-validation results.

Generalization failures arise from well-known issues: dataset shift, spurious correlations, confounding, and data leakage(2; 4). While proposed solutions exist, there is still no clear, practical method to answer the key question: will this model work for new patients? This portfolio directly addresses this challenge by presenting a three-stage, actionable framework for pre-deployment diagnostics that enables performance estimates without specialized resources.

II Critical Literature Review

II-A Surveying the Field

The basic reasons models fail to perform well on new groups are well known. Ben-David et al. (4) showed mathematically that a model's ability to generalize across groups depends on how different those groups are. After a certain point, no method, no matter how advanced, can fix this difference. Gretton et al. (6) built on this by developing the Maximum Mean Discrepancy (MMD) test, a measure of how different two sets of data are. The key point is that you cannot fix data differences unless you first identify them.

These theoretical limitations manifest in various real-world contexts. In genomics, Whalen et al. (2) identified five common errors that can artificially inflate model performance: non-independent data splits, test data leakage into training sets, overlooked confounding factors, unbalanced class distributions, and test data that does not reflect real-world application. In clinical biomarker research, Wörheide et al. (5) distinguished between covariate shift, where input features change, and prior probability shift, where disease prevalence changes. They demonstrated that conflating these issues leads to inappropriate corrective measures. Importance weighting, a commonly used method, was found to benefit simple models but to impair the performance of more flexible models such as gradient boosting. Distributionally robust optimization (DRO) was highlighted as a promising approach by Duchi and Namkoong (7) and Rahimian and Mehrotra and (8), although its effectiveness was not empirically evaluated in these studies.

In immunology, Moris et al., 2021 (9) demonstrated that models estimating T-cell receptor binding performed well on targets encountered during training but failed entirely on novel targets, with results equivalent to random chance. This issue was obscured because aggregate performance metrics were dominated by well-represented groups. Chekroud et al.(3) illustrated a similar phenomenon: schizophrenia treatment models that appeared highly accurate in individual clinical trials did not generalize effectively across multiple trials.

II-B Assessment of Proposed Solutions and Associated Limitations

The literature offers several answers to the problem of generalization. Ektefaie et al. (10) introduced SPECTRA, a system for testing models across different levels of overlap between training and test data, and found that all 19 deep learning models tested performed worse as overlap decreased. Oloruntoba et al. (11) showed that using realistic data splits yields much lower performance than the usual random splits, suggesting that common evaluation methods are overly optimistic. Pavlovic et al. (12) noted that the main problem is that models learn spurious associations rather than true cause-and-effect links and showed that selecting features based on causal graphs yields more reliable results. However, their method needs experts to build these graphs beforehand, which is rarely possible.

Arjovsky et al. (13) introduced invariant risk minimization (IRM), which aims to find features that reliably predict outcomes across different training settings rather than just one. Although IRM is strong in theory, it is hard to apply in practice because it requires multiple distinct training environments, which are rarely available in clinical data. To address reporting standards, Collins et al. (1) developed the 27-item TRIPOD+AI checklist to improve consistency in documenting prediction model studies. While this checklist improves clarity, it does not guarantee quality; a model can meet all checklist points but still fail when used. Studies by Lever et al.(14), Okser et al. (15), Runcie and Cheng (16), Dincer et al. (17), Evans et al. (18), and Steyerberg et al. (14) highlight specific examples of a common problem: standard evaluation methods commonly overestimate how well models perform.

II-C Conceptual Frameworks for Generalization Failure and Their Limitations

To provide context for these findings, the literature examines generalization failure from three main viewpoints, each with distinct limitations.

The first view sees the problem as a methodological one. (2) list avoidable mistakes, (1) set reporting rules, (19) find the right sample sizes, and (20) cover ethical duties. The main idea is that meticulous methods lead to models that work well in general. While this advice makes sense, it is not sufficient, as failures still occur even when methods are properly applied, as shown by (3) across five well-designed studies, reinforcing concerns raised in earlier paragraphs.

The second view sees the problem as one of data, pointing out that the groups used for training and for real use often differ significantly, and the main challenge is handling this difference. (4) demonstrated mathematically that some population differences cannot be overcome. (5) found that fixes depend a lot on the model and are often hard to predict. Moris et al. (2021) (9) observed that completely new inputs can lead to total model failure. (21) argue that the only lasting fix is to find features that truly cause outcomes, not just those linked by chance. This debate echoes the conflict seen in previous practical and theoretical interpretations, and persists because no direct comparisons using the same data have been conducted.

The third and most fundamental perspective locates the issue within the principles of machine learning. (13) and (12) argue that typical machine learning approaches prioritize minimizing training error, leading models to memorize the idiosyncrasies of the training population and failing to generalize. Both causal modeling and invariant risk minimization attempt to address these flaws but encounter practical barriers. Watson and Szathmary (2016) (22) further relate generalization deficiencies to the transfer of adaptations between environments in biological evolution.

Freiesleben and Grote (2023) (23) highlight an important distinction: statistical generalization means a model performs well on test data from the same group, while practical robustness means it continues to perform well when the group changes. Cross-validation checks statistical generalization, but real-world use needs practical robustness, a distinction often blurred in research, further complicating efforts discussed previously.

(24) outline the standards for good performance when models are used, including discrimination, calibration, and usefulness in clinical settings. (3) show how often real models fail to meet these standards, and (1) ensure that these failures are carefully recorded. Despite this, the research mostly ends with a description of the problems, echoing the limited predictive tools highlighted earlier. While the field has strong tools to explain generalization failures after they occur, it has few ways to predict them beforehand.

II-D Comprehensive Critical Analysis and Synthesis

II-D1 Foundational Literature

Choose Chekroud et al. (2024) (3) Science because it is recent, respected, and uses strong evidence from five international clinical trials. Performance drops sharply from 0.72 during the trial to 0.54 in a new trial, across four definitions of treatment success and two algorithms. These results cannot be ignored. The main limitation is focus: predicting schizophrenia treatment response is very hard. Saying ”these models fail” and suggesting ”clinical models generally fail” is not fully fair. The study does not separate fitting random noise from problems with applying models to new data, which need different fixes.

Without testing ways to improve reliability, the authors leave open the question of whether the problem can be solved.

Whalen et al. (2022) (2), published in Nature Reviews Genetics, are notable for their clear, practical approach and reproducible notebooks. Their five-pitfall framework is clear: using related examples can improve performance by over 0.5, while poor data preparation can make models look accurate even when there is no real signal. Unlike Chekroud et al., Whalen et al. assume models work once technical mistakes are fixed. Chekroud et al.'s models fail for deeper reasons because statistical patterns break down across trials. Also, Whalen et al.'s approach to handling confounders is weak; removing confounders via principal component analysis does not ensure that learned relationships generalize.

Building on these methods, Pavlovic et al. (2024) (12) offer the most practical solution in the literature in Nature Machine Intelligence. Their causal graph method identifies truly predictive features rather than confounded ones. Simulations show that confounders cause double errors; selection bias makes models no better than chance on unbiased data. Limitation: experiments assume perfect knowledge of cause and effect, which is impossible in real clinical settings. It is still unclear whether rough causal graphs actually work better than standard models or make errors worse.

Extending the study of model generalization, Moris et al. (2021) (9) in Briefings in Bioinformatics test the limits by predicting outcomes for biological targets completely missing from the training data. Their main finding is the decoy experiment: published models that appeared to predict T-cell receptor binding performed just as well when epitopes were replaced with random sequences, suggesting they memorized receptor patterns rather than learning real biological relationships. Their own model shows some ability to predict beyond training data, but only for epitopes very similar to the training data, not true generalization.

Turning to reporting standards, Collins et al. (2024) (1), TRIPOD+AI, and BMJ offer trusted guidelines developed by 200 participants. The 27-item checklist adds fairness and open science, but it is a tool for transparency—not for ensuring quality. A model that follows the checklist can still fail with new patients. Item 26 asks for a discussion on generalizability but does not require enough detail. A shallow statement is treated the same as a thorough analysis. If standards cannot tell good from poor reporting, they cannot stop failures when models are used.

II-E Synthesis of Findings

The literature converges on three points: internal validation alone does not establish generalization; dataset shift occurs in nearly all real-world deployments, yet there is no consensus on mitigation strategies; and spurious correlations cannot be addressed solely with standard statistical techniques.

Two major disagreements persist. The first concerns whether generalization failure is primarily a data quality issue, implying that methodological improvements will resolve it, or a structural issue, in which case even optimal methodology cannot prevent failure if the model learns incorrect patterns. Chekroud et al. (3) employ standard methods and observe failure without testing causal approaches, while Pavlovic et al. (12) demonstrate the benefits of causal methods but do not compare them to well-implemented standard corrections. The definitive experiment to resolve this debate has not yet been conducted. The second disagreement contrasts the practical optimism of applied researchers, who believe methodological corrections can succeed, with the theoretical caution of Ben-David et al. (4), who demonstrate that beyond a certain threshold of distributional difference, no correction is effective. Currently, the field lacks a means to determine which scenario applies in specific deployments.

Three major gaps stand out: no current method can predict before use if a model will fail in a new setting; monitoring model performance after deployment is not a main research focus; and the best solution, causal modeling, needs expert skills that most clinical teams do not have. The field needs a practical system that estimates performance problems before use, identifies which features are reliable, provides clear guidance, and enables tracking performance after deployment, all without requiring causal graphs or expert teams. This is a problem of putting pieces together, not just knowing about it, and this project aims to fix it.

III Research Gap and Objectives

Existing literature provides robust methodologies for analyzing generalization failure; however, it lacks predictive tools applicable prior to deployment.

The main problem is not a lack of awareness, since the issue is well covered in existing studies. Instead, the problem is the lack of a combined process that includes measuring data differences, analyzing individual features, and providing deployment advice.

Research Question:

How well can we predict before deployment if clinical machine learning models will perform worse on new patient groups by checking data differences and feature reliability? Also, what cutoff points can be set to decide if it is safe to deploy the model?

Objectives:

1) **Objective 1:** Apply MMD (6), KL divergence, and Wasserstein distance methods (7; 8) to clinical data from different settings where performance drops are known. Compare each score with the real loss in balanced accuracy.
2) **Objective 2:** Use leave-one-context-out permutation importance (2) to mark each predictor as either stable or sensitive to changes. Check these labels against actual performance across different settings, aiming for at least 70% agreement..
3) **Objective 3:** Combine Objectives 1 and 2 into a decision flowchart with four options: deploy with normal monitoring, deploy with extra monitoring, retrain using stable

features, or do not deploy. Test this on at least one separate setting and clearly report its limitations.

IV Project Scope and Methodology

IV-A Scope Definition

This project creates a tool to evaluate clinical prediction models before they are used, estimate how their accuracy might drop when applied to a new group of patients, and suggest whether to use, change, or delay the model. It does not try to fix problems with the model working in new settings; advanced methods like causal modeling and domain adaptation are still too hard for most teams. The main goal is to find risks before any harm happens. The tool works with organized clinical data that includes clear subgroups such as hospitals, regions, trial sites, or time periods, and only requires basic Python skills.

IV-B Three-Stage Architecture and Technical Pipeline

Figure 2 provides an overview of the complete diagnostic pipeline. Each stage answers one question, produces one output, and feeds into the next.

a) Stage 1 - Divergence Scoring:
How different is the new group from the training group? Three measures are calculated at the same time:

- **MMD** (6), which compares data without assuming its shape
- **KL divergence** (7) (via Wörheide et al. (5)), which finds when the new group has patients not seen in the training data; and
- **Wasserstein distance**(via Wörheide et al. (5)), which works better than KL when the groups have little overlap.

Wörheide et al. list all three as common choices but never check which one best predicts real drops in performance. This project addresses that directly.

b) Stage 2 - Feature Stability:
Which features can be trusted in different situations? Using a method that excludes one context at a time, each feature is scored for consistency across groups and labelled shift-stable (> 0.6), uncertain (0.3–0.6), or shift-sensitive (< 0.3). This is similar to what Pavlovic et al. (12) do with causal graphs: they identify features that maintain their link to the outcome across environments without requiring anyone to create a causal diagram, which makes their method hard for most teams to use.

c) Stage 3 - Decision Framework:
What should the practitioner do? Divergence score and stability profile combine into four possible recommendations, as shown in Table I.

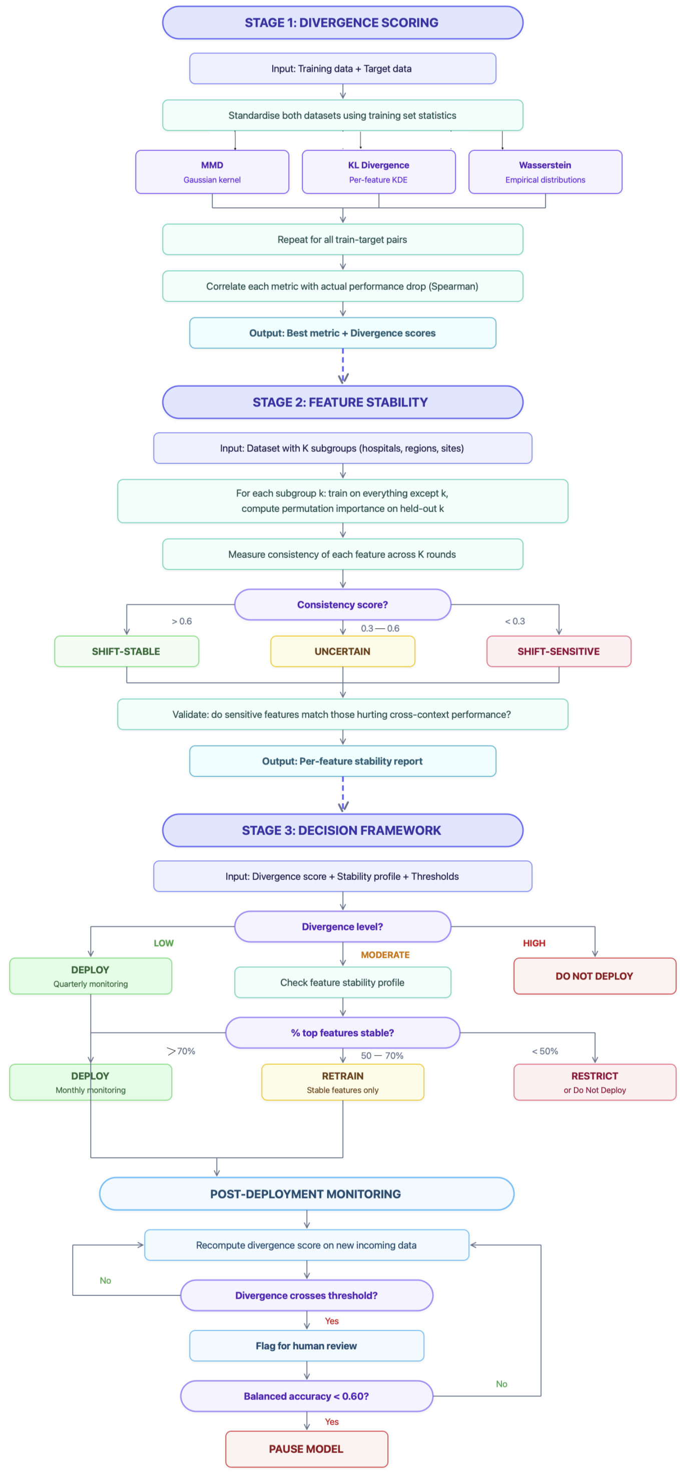

Fig. 1: Three-stage pre-deployment diagnostic pipeline. Stage 1 computes distributional divergence between training and target data. Stage 2 evaluates feature stability via leave-one-context-out permutation importance. Stage 3 combines both outputs into a deployment recommendation.

TABLE I: Deployment Decision Matrix

Divergence	Feature Stability	Recommendation
Low	–	Deploy, quarterly monitoring
Moderate	≥70% stable	Deploy, monthly monitoring
Moderate	50–70% stable	Retrain on stable features only
Moderate/ High	<50% stable	Restrict or do not deploy
High	–	Do not deploy

Post-deployment, divergence is recomputed on incoming data at regular intervals. If it exceeds the upper threshold, the model is flagged for human review; if balanced accuracy falls below 0.60, the cross-validation baseline set by Chekroud et al. (3), the model is paused.

IV-C Baselines and Justification

The framework is compared with three baseline methods, each representing the best current approach from a different area of the literature under review.

Baseline 1: Naive cross-validation. This is how most models are currently evaluated: internal cross-validation performance is treated as the expected deployment performance. Chekroud et al. (3) show that this approach fails badly (0.60 on internal data versus 0.54 on independent data). If the proposed framework predicts the actual performance drop more accurately than naive CV does, it adds value.

Baseline 2: Importance weighting. This is the best single correction method tested in the reviewed research, according to Wörheide et al. (5). It works well for simple linear models, but worsens results for flexible models like gradient boosting. The comparison checks whether diagnosing the problem (the new approach) works better than fixing it directly with an existing method.

Baseline 3: Random feature subsets. Models are retrained using randomly selected features, with the same number as in the stable feature set. This tests whether the stability-based feature selection in Stage 2 really finds better features or if any group of the same size would work just as well.

IV-D Evaluation Metrics

The main measure is the Spearman correlation between the predicted performance drop (from Stage 1 divergence scores) and the actual performance drop seen on external data. This directly tests if measuring data differences before deployment can predict how much performance will change, which is the framework's main claim.

Secondary metrics include:

- Balanced accuracy (matching Chekroud et al.'s primary measure) (3);
- Calibration slope (following Steyerberg et al. (24) and the TRIPOD+AI calibration requirements from Collins et al. (1));
- Balanced accuracy broken down by demographic subgroups (following the TRIPOD+AI fairness items);
- Concordance rate between predicted and observed feature instability (target: at least 70%, as set out in Objective 2).

V Data, Ethics, and Practical Considerations

V-A Data Readiness

The primary dataset is the eICU Collaborative Research Database Demo (25), a freely available ICU subset hosted on PhysioNet containing 2,520 patients across 186 US hospitals. Variables include demographics, APACHE physiology scores, lab results, diagnoses, and in-hospital mortality. The multi-hospital structure is what makes it suitable, as each hospital or region represents an independent clinical environment with its own patient mix, providing exactly the cross-context variation the framework requires.

The demo subset can be downloaded freely without special access; the full database (73,718 patients) requires CITI training and a signed data use agreement. The PhysioNet license allows research use but not sharing or commercial use. Existing standard processes from the YAIB framework (26) handle data handling and basic modeling, allowing the project to focus on the diagnostic framework.

According to the Lawrence Framework scale, the data is at DRL-3, organized but requiring specific preparation steps, such as combining CSV files, filling in missing data, and adjusting age records. After this preparation, it reaches DRL-5. There are three limits: all hospitals are in the US, which limits how well results apply elsewhere; the demo has only 10–40 patients per hospital, so hospitals are grouped into four US regions (Midwest, South, West, Northeast) with 140–807 patients each; and 80% of the patients are Caucasian, mostly older and male, which limits fairness testing across different groups.

V-B Ethical Considerations

Fairness. The decision limits are set using data mostly from one group 80% Caucasian, 9.2% African American, 5% Hispanic, 1.2% Asian. These limits might not work well for groups with different mixes. The smallest region (Northeast, 159 patients) has the highest death rate (11.9% compared to 7.1% in the Midwest), showing real differences between areas. A system made to point when models don't work well everywhere might wrongly reassure the groups most likely to be left out.

Privacy. The data has personal details removed, but some demo hospitals have as few as 10 patients. Calculating statistics on small groups carries a small risk that individuals could still be identified, especially if combined with other information. Under UK GDPR, data without names can still count as personal data if people can realistically be identified.

Automation bias. The system gives clear choices: deploy, retrain, restrict, or do not deploy. There is a real risk that people see these as certain decisions rather than estimates, which could lead to risky use of borderline models or to stopping useful ones. Also, a low difference score for a whole group does not mean every single patient in that group is well represented in the training data.

Data governance. The PhysioNet license requires that work stay within agreed-upon rules. In a UK clinical setting, the DUAA 2025 would require that patients have the right to a human review of any automated decision affecting them, and that the reasons be clear and verifiable.

V-C Mitigations

TABLE II: Ethical Risks and Mitigations

Risk	**Mitigation**
Unfair thresholds	Report all results stratified by demographic subgroup; explicitly state calibration population limitations
Re-identification	No subgroup statistics below $n = 10$; divergence computed on aggregate distributions only; no data linkage
Automation bias	Display divergence score and uncertainty range alongside categorical recommendation; require documented human decision before any deployment action
Feature exclusion inequity	Before dropping shift-sensitive features, check whether removal disproportionately harms specific demographic groups; use targeted recalibration if so
Governance breach	Follow all PhysioNet data use terms; if deployed in UK clinical settings, implement DUAA 2025 human-review and auditability requirements

VI Project Plan

The project plan is presented in two parts: what was actually completed over 8 weeks, and a hypothetical 12-week lifecycle showing how the full project would run if repeated with the benefit of hindsight.

In reality, the literature review and critical analysis took about 5 weeks, rather than the 3 weeks a more experienced researcher might need. Learning about causal modeling and distributional shift theory for the first time meant reading the papers several times and going back and forth before the ideas made sense. Report writing began in Week 2 and continued throughout, rather than being left until the end, which helped keep the argument clear as the analysis progressed. Methodology, data preparation, and implementation occurred concurrently during Weeks 6 and 7, and the final portfolio was submitted in Week 8.

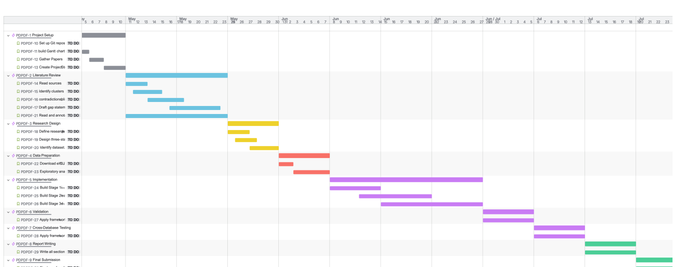

Fig. 2: Hypothetical Project Plan

The 12-week plan shortens the literature phase to 3 weeks and adds validation and testing on different databases that the 8-week schedule could not include. The process follows a clear order. Weeks 1 to 3 cover the literature review and critical analysis, which must be completed before the research design in Week 4, as the method cannot be planned without understanding where current approaches fall short. Data preparation in Week 5 depends on the dataset chosen during the research design. Implementation happens in Weeks 6 to 8 in a strict order. Stage 1 divergence scoring must finish before Stage 2 feature stability starts, since Stage 2 needs the trained models from Stage 1. Stage 3 cannot set its thresholds or create the decision flowchart without results from both earlier stages. Validation in Weeks 9 and 10 depends on having the full working system, testing first on the full eICU database to see if demo-based thresholds work at scale, then on MIMIC-IV as a truly independent external database. Report writing gets a full week (Week 11) instead of being squeezed into the last days, using notes and results from Weeks 6 to 10. Week 12 is for final review, formatting, and submission. The biggest risk is the move from literature to design. If the gap statement is not clear by the end of Week 3, everything after that will be delayed.

Tasks and milestones are tracked across two Jira boards. The actual project board documenting the completed work is available at Jira board (alternative: Git link). The hypothetical 12-week plan is organized on a separate board at hypothetical Jira board (alternative: Git link), with each phase in a dedicated column and task cards progressing from To Do through In Progress to Done.

VII Non-Technical Summary

When hospitals use computer programs to predict patient outcomes, like who might respond well to a treatment or who is most likely to get worse, those programs are usually tested on data from the same hospital or study that made them. The problem is that patients at a different hospital, in another part of the country, or from a different background may not be similar to the patients the program learned from. When that happens, the program's predictions can become unreliable, sometimes no better than a random guess.

Research published in *Science* in 2024 showed this clearly: models that seemed highly accurate in their original clinical trials performed at coin-flip levels when tested on separate datasets (3). This is not a rare or unusual failure; similar patterns have been found in genetics, immune system studies, and biomarker research. The tools to explain why this happens already exist. There is no practical way to check, before a program is used on real patients, whether it is likely to work in that specific setting.

This project fills that gap. The idea is simple: before using a prediction program on a new group of patients, measure how different those patients are from the ones the program was trained on. If the difference is small, the program is probably safe to use. If the difference is large, it probably is not. The system also checks which pieces of information the program depends on; some work well across different patient groups, while others apply only to the specific group the program was built for.

The result is a simple scoring system and decision chart that tells doctors and data experts whether to proceed, make changes, or wait. Early testing on real ICU data from 186 US hospitals shows the method can spot risky situations. It does not fix every problem in medical AI, but it tackles the most dangerous one: using a tool that looks good on paper without knowing if it will actually help the patients you are treating.

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